

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

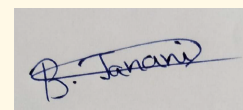
ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	Total Marks:	20 marks
CO1 Statement:	Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.		
Student Name:	B Janani	Reg. No.:	192321100
Section / Batch:	Slot A- 2023	Date:	22.07.2026

Code of Conduct

I B JANANI (192321100) (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Instructions

- This test contains ONE question.
- Duration: 30 minutes.

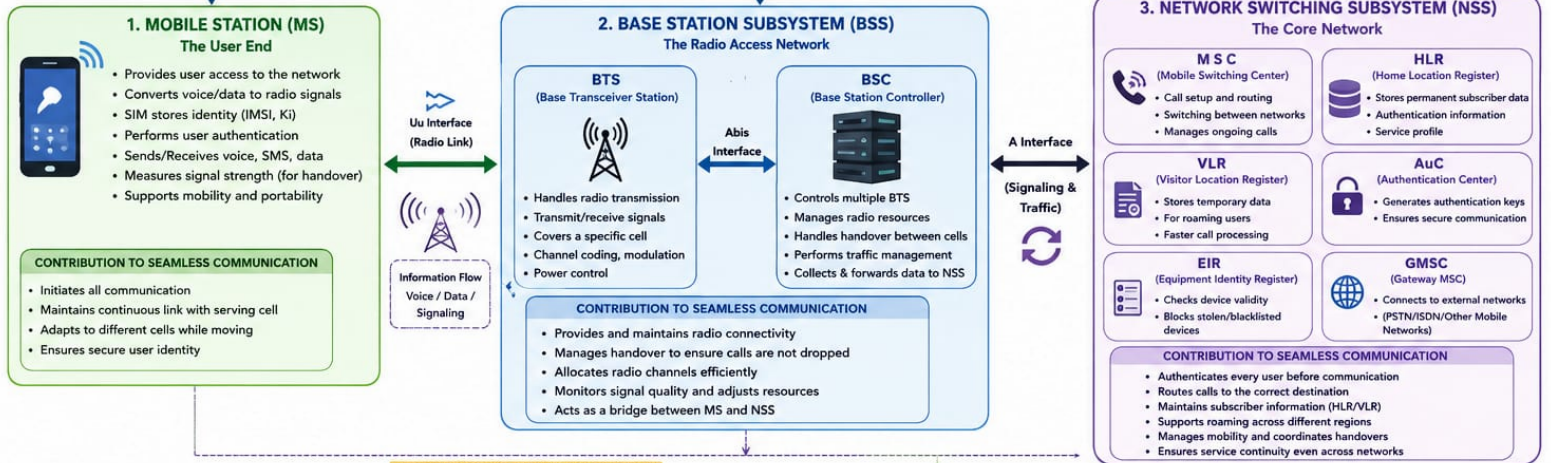
Annexure A - QUESTIONS

1. Construct a concept map that evaluates the contribution of Mobile Station, Base Station Subsystem, and Network Switching Subsystem toward seamless communication.
Illustrate their interdependencies.
Assess which subsystem has the greatest influence on service continuity.
Justify your conclusions using concept relationships.

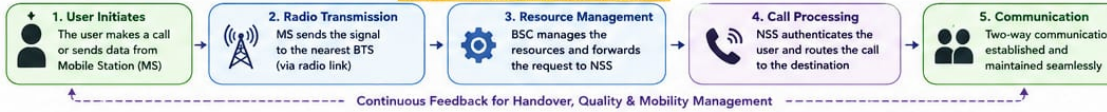
CONCEPT MAP: CONTRIBUTION OF GSM SUBSYSTEMS TOWARD SEAMLESS COMMUNICATION

SEAMLESS & CONTINUOUS COMMUNICATION

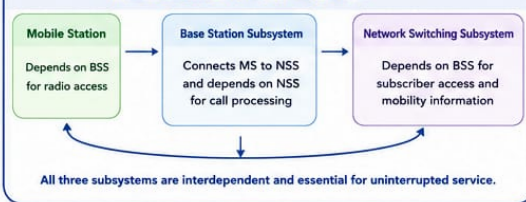
Reliable · Uninterrupted · Secure · Anywhere · Anytime



HOW THE SUBSYSTEMS WORK TOGETHER



INTERDEPENDENCIES AT A GLANCE

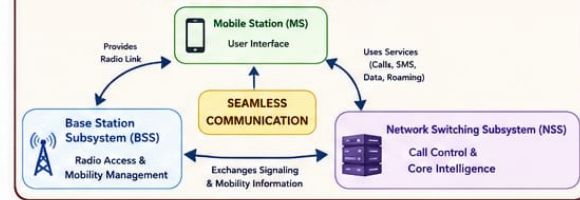


ASSESSMENT: WHICH SUBSYSTEM HAS THE GREATEST INFLUENCE ON SERVICE CONTINUITY?

NETWORK SWITCHING SUBSYSTEM (NSS) HAS THE GREATEST INFLUENCE.

- REASONS (BASED ON CONCEPT RELATIONSHIPS)
- ✓ Without NSS, calls cannot be authenticated or routed.
 - ✓ Maintains subscriber databases (HLR/VLR) that enable service.
 - ✓ Manages roaming, call forwarding, and inter-network connectivity.
 - ✓ Coordinates with BSS during handover to update user location.
 - ✓ Ensures service continuity even when user moves across regions or networks.
 - ✓ Acts as the brain of the network that controls and supervises the entire communication process.

CONCEPT RELATIONSHIPS EXPLAINED



FINAL CONCLUSION

Mobile Station enables user communication, Base Station Subsystem provides the radio connection and manages handovers, and Network Switching Subsystem controls authentication, routing, mobility, and databases. While all are essential, the NSS has the greatest influence on service continuity because it ensures that communication is authenticated, routed, and maintained seamlessly across time, location, and networks.

KEY TAKEAWAY

Seamless communication is the result of the strong interconnection and cooperation among MS, BSS, and NSS, with NSS acting as the central coordinator that guarantees uninterrupted and reliable service.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____